

EX NO: 2a	GENERATION AND ANALYSIS OF FREQUENCY MODULATED (FM) SIGNALS USING MATLAB
DATE:	

CODE

```

clc;
clear;
close all
% Parameters
Am = 1;
Ac = 1;
fm = 5;
fc = 50;
beta = 5;
fs = 1000;
t = 0:1/fs:1;
% Message Signal
m = Am*cos(2*pi*fm*t);
% Carrier Signal
c = Ac*cos(2*pi*fc*t);
% FM Signal
fm_sig = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
% Frequency Spectrum
N = length(fm_sig);
f = linspace(-fs/2, fs/2, N);
FM_FFT = abs(fftshift(fft(fm_sig)))/N;
% Plotting
figure('Name','Frequency Modulation','NumberTitle','off')
subplot(4,1,1);
plot(t,m,'b');

```

```

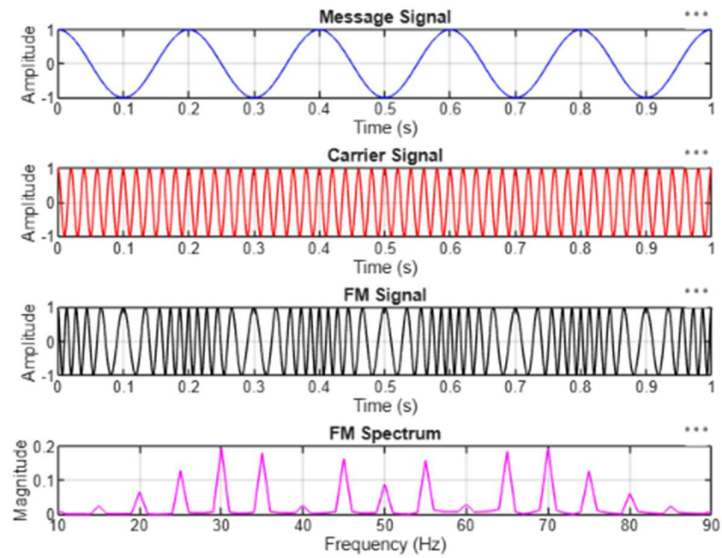
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(4,1,2);
plot(t,c,'r');
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(4,1,3);
plot(t,fm_sig,'k');
title('FM Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(4,1,4);
plot(f,FM_FFT,'m');
xlim([fc-40 fc+40]);
title('FM Spectrum');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;
% Display Results
fprintf('Carrier Frequency (fc) = %d Hz\n', fc);
fprintf('Message Frequency (fm) = %d Hz\n', fm);
fprintf('Modulation Index ( $\beta$ ) = %.2f\n', beta);
% Carson's Rule Bandwidth
BW = 2*(beta*fm + fm);

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fprintf('Approximate FM Bandwidth = %.2f Hz\n', BW);
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OUT PUT :

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Carrier Frequency (f_c) = 50 Hz

Message Frequency (f_m) = 5 Hz

Modulation Index (β) = 5.00